

POST-QUANTUM CRYPTOGRAPHY (PQC) PLATFORM

Handshake Architecture, Session Key Derivation & Recent Real-World Cyber Attacks Guide

PART 1: Handshake & Session Key Deep-Dive (Crystal Clear Explanation)

1. Handshake-na Enna? Simple Real-World Analogy:

- Rendu per (User A & User B) secret-ah chat panna poraanga. Adhukku rendu per kittayum ore **Secret Lock Key (AES Session Key)** irukanum. Aana indha key-ai internet moolama hacker paakaama epdi exchange panradhu? Idhukku peyar than **Cryptographic Handshake**.

2. ML-KEM Encapsulation vs Decapsulation (Epdi Work Aagudhu?):

- Step 1 (User A generates Public Lockbox):** User A oru magic open lockbox (ML-KEM Public Key) create panni internet-la User B-kku anupuvaaru. User A kitta mattum than unlock panna Private Key irukum.
- Step 2 (User B Encapsulates Secret):** User B oru random secret key (*kem_secret*) create panni, User A-oda lockbox-kulla pottu lock panniduvaaru. Indha locked box peyar than *kem_cipher*. Idhukku peyar **Encapsulation**.
- Step 3 (User A Decapsulates):** Locked box User A-kitta vandhadhum, avar thannoda Private Key pottu open panni *kem_secret*-ai eduthukkuvaaru. Idhukku peyar **Decapsulation**. Ipo rendu per kittayum ore secret irukku!

3. Hybrid Mode (Dual Locks - X25519 + ML-KEM):

- Oru lockbox-ku badhila rendu locks podrom: (1) Classical Elliptic Curve Lock (X25519) + (2) Post-Quantum Lattice Lock (ML-KEM-768). Rendu lock-um serndha than *dh_secret* + *kem_secret* kedaikum.
- Why?** Oruvela future-la PQC lattice math-la edhavadhu flaw kandupidichalum, classical 128-bit ECC data-va protect pannum (Defense-in-Depth).

4. HKDF-SHA256 (Session Key Creation):

- Rendu perum kedaicha (*dh_secret* || *kem_secret*)-ai salt and context string ("pqc-secure-platform-hybrid-v1") kooda serthu **HKDF-SHA256** cryptographic blender-la poduvanga. Idhula irundhu identical **256-bit AES Symmetric Session Key** create aagum. Message bulk encryption idhula than nadakkum.

Handshake Presentation Speech (Tanglish):

'Handshake-na oru safety box maathiri. User A oru open PQC lockbox anupuraaru. User B thannoda secret key-ai ulla vechu lock panni (Encapsulation) thirumba anupuraaru. User A thannoda private key vechu open panraaru (Decapsulation). Classical X25519 + Quantum ML-KEM rendu key-aiyum HKDF-la merge panni 256-bit Session Key generate panrom. Idhanaala end-to-end hacker-aal oru bit kooda thiruda mudiyaadhu!'

PART 2: Recent Real-World Cyber Attacks & How PQC Solves Them

Attack Scenario & Real-World Incident	How Attackers Exploit Classical Systems	How Our PQC Platform Solves It
1. Telecom Wiretapping & Subsea Taps (e.g., Salt Typhoon / Nation-State Taps on AT&T, Verizon & Subsea Cables)	Hackers tap fiber optic lines and intercept petabytes of encrypted TLS/IPsec data. They store it today for HNDL (Harvest Now Decrypt Later) to crack with quantum computers.	ML-KEM Lattice Encapsulation: Mathematical shortest-vector problem (SVP) in high-dimensional lattices cannot be solved by Shor's algorithm. Stored data remains uncrackable forever.
2. Man-in-the-Middle (MITM) & Spoofing (e.g., DNS/BGP Hijacking & Fake Certificate Authorities)	Attackers forge RSA-2048 or ECDSA digital signatures using quantum period-finding algorithms, impersonating legitimate banking servers or users.	ML-DSA-65 & SLH-DSA: Post-quantum zero-trust digital signatures sign every handshake packet. Quantum computers cannot forge lattice or hash-based signatures.
3. Financial Replay Attacks (e.g., SWIFT & Interbank Wire Transfer Spoofing)	Adversaries capture valid encrypted money-transfer packets and retransmit them multiple times to drain accounts or manipulate balances.	Monotonic Sequence + AES-GCM AAD: Every message increments sequence (seq=1, 2...). Duplicate sequence triggers immediate ATTACK_REPLAY error; GCM tag detects 1-bit changes.
4. Lifetime Genomic & Medical Data Theft (e.g., 23andMe / Healthcare Data Breaches)	DNA sequencing and personal medical data never change over a person's entire lifetime (50-80 years). Decrypting it 10 years later still causes catastrophic privacy loss.	Permanent Forward Secrecy: Ephemeral PQC keys + FIPS 203 standards ensure healthcare records have permanent mathematical confidentiality exceeding 50+ years.
5. Critical Infrastructure / SCADA Sabotage (e.g., Power Grid & Water Utility Attacks)	IoT microcontrollers have low CPU power, so they use weak RSA/ECC or no encryption at all, making remote telemetry commands vulnerable to quantum cyberwarfare.	Ascon-128a (NIST LWC): Ultra-lightweight authenticated encryption designed specifically for resource-constrained IoT edge devices, sensors, and power grids.

PART 3: Global Regulatory Mandate (CNSA 2.0 & CISA Timeline)

- US NSA & CISA CNSA 2.0 Mandate:** All national security systems, military, and critical infrastructure **MUST** begin PQC migration by 2025 and achieve 100% exclusive PQC enforcement by 2030–2033.

- **NIST FIPS Release (Aug 2024):** NIST officially released FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA). Our platform is built 100% directly upon these official standards.
- **Commercial Feasibility:** Enterprise-ready at ~\$1,045/month for 10k-50k users, providing bank-grade security at startup cloud pricing.

■■ **Real-World Attacks Presentation Speech (Tanglish):**
'Recent-ah nadandha Salt Typhoon telecom breach and undersea wiretapping paatha, hackers data-va thirudi store panranga (HNDL). Innaiki RSA/ECC use panna 5-10 varushathula ellam open aagidum. Namma platform ML-KEM lattice encryption, ML-DSA signatures and Ascon-128a IoT security moolama Telecom, Defense, Banking, and Medical data-kku 100% unbreakable quantum protection tharudhu. NSA CNSA 2.0 mandate-kku namma platform perfect solution!'